

DRUGS ACTING ON THE CENTRAL NERVOUS SYSTEM

BIETE LUNDAU LUKE

DipPharm,Bpharm,Mclinpharm

OVERVIEW OF THE CNS

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- ▶ The CNS consists of the brain and the spinal cord
- ▶ Sensory information arrives to the CNS from the special senses and peripheral nerves and when integrated with memories and internal drive states, cognitive, emotional and motor (behavioral) responses are generated
- ▶ This process is enhanced by the complex interplay of neurotransmitters and neuromodulators acting on their receptors to excite or inhibit CNS neurons
- ▶ Most CNS drugs correct an imbalance in neurotransmitter or their receptor and are thus used to relieve symptoms of brain dysfunction though do not usually correct the underlying disorder
- ▶ A neuromodulator is a general term for any substance that exerts an effect on neurotransmission among a set of neurons in the brain

REMEMBER: *“The CNS coordinates and controls the activity of the peripheral nervous system”*

Neurotransmitters and receptors

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- ▶ CNS neurotransmitters can either evoke an excitatory or an inhibitory synaptic membrane and trigger effects at pre- and post – synaptic sites on target neurons
- ▶ Neurotransmitters in the CNS include acetylcholine, several amino acids, biogenic amines and neuropeptide
- ▶ **Acetylcholine** acts as excitatory or inhibitory neurotransmitter in a number of neuronal tracts
- ▶ Amino acids involve **γ-amino butyric acid (GABA)**, **glycine**, **glutamate** and **aspartate**
- ▶ GABA and glycine are inhibitory neurotransmitters and most drugs affecting GABA primarily activate or inhibit the GABA
- ▶ **Glutamate** and **aspartate** are acidic amino acids that function as excitatory neurotransmitters in the CNS

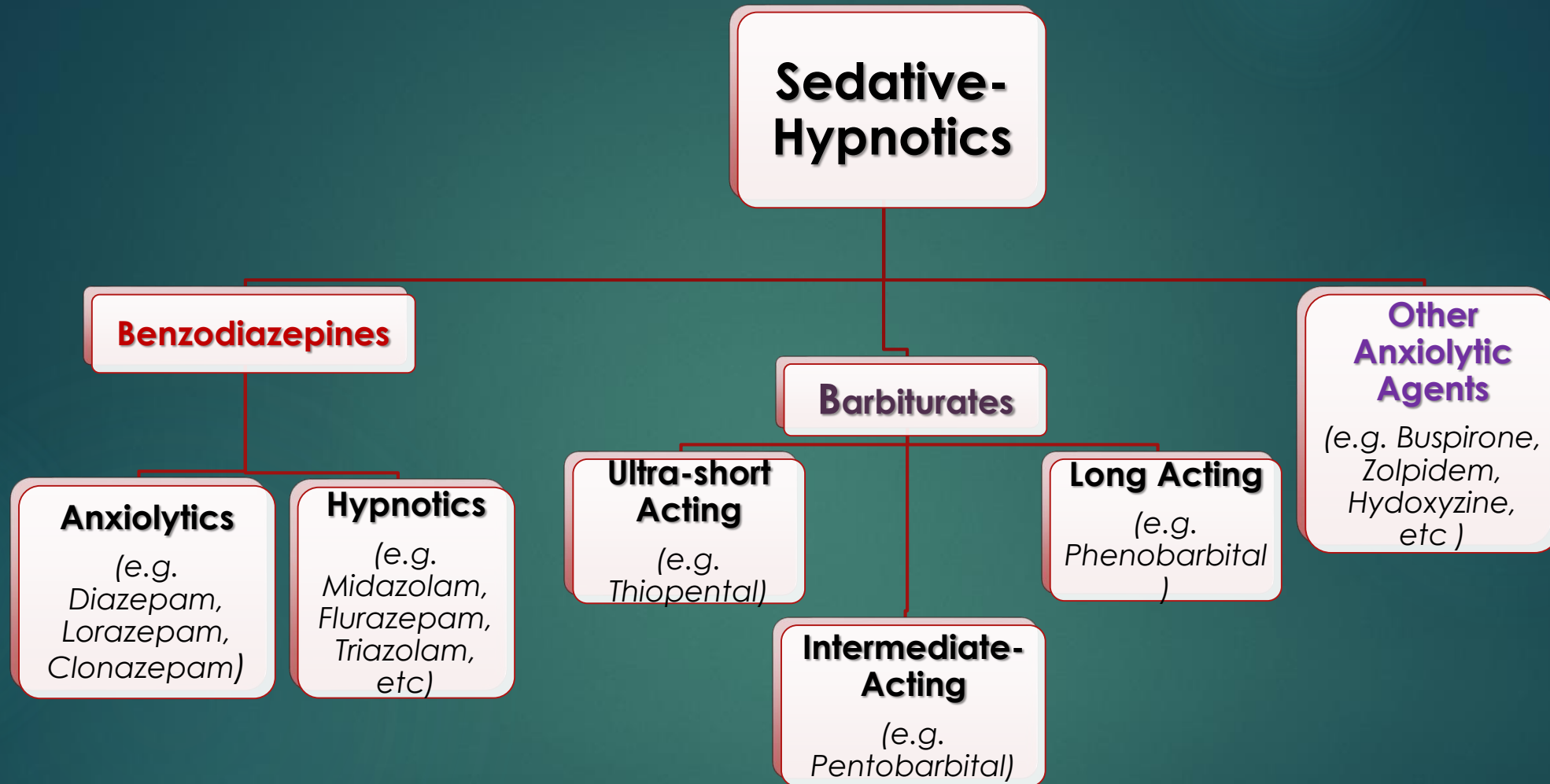
CNS Neurotransmitters

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1. Gamma aminobutyric acid (GABA)
2. Glycine
3. Aspartate
4. Glutamate
5. Acetylcholine
6. Catecholamines (*i.e.* Dopamine, Norepinephrine and Epinephrine).
5. 5-Hydroxytryptamine (5-HT, Serotonin)
6. Histamine
7. Purines (*e.g.* Adenosine)
8. Endogenous peptides (*e.g.* Opioid peptides, Substance-P, etc)

SEDATIVE – HYPNOTIC AND ANXIOLYTIC DRUGS

- ▶ Sedative – hypnotic drugs are among the most widely used pharmaceutical agents in the world
- ▶ The sedative part of the name refers to the ability of these agents to calm or reduce anxiety, also known as anxiolytic effects
- ▶ The hypnotic part of the name refers to the ability of these agents to induce drowsiness and promote sleep
- ▶ Hypnosis is caused by a greater depression of the CNS which mostly is achieved at higher doses while sedation at relatively lower doses
- ▶ A few agents exert anxiolytic effects without necessarily causing sedation or hypnosis



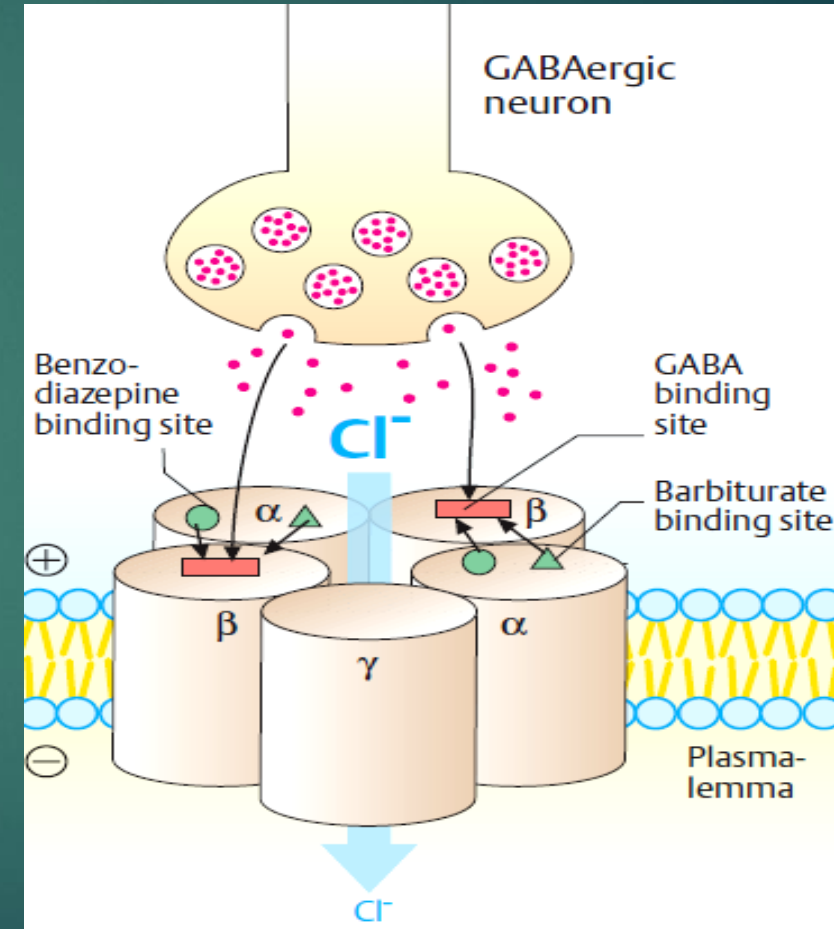
Mechanism of action

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They bind to and potentiate GABA receptor activity & trigger opening of Cl^- ion channels:

→ hyperpolarization of postsynaptic potentials

→ inhibit firing of action Potential



Luellmann, 2005

Benzodiazepines:

Pharmacological effects

- ▶ Anxiolytic (*at low doses*)
- ▶ Sedation & Hypnosis (*at high doses*)
- ▶ Anticonvulsant
- ▶ Skeletal muscle relaxation
- ▶ Euphoria
- ▶ Anaesthetic action

Classification of Benzodiazepines:

1. Long acting BZD:

- **Diazepam:** - Anxiolytic, Anticonvulsant, Skeletal muscle relaxant
- **Clonazepam:** - Anxiolytic and Anticonvulsant

2. Intermediate acting BZD:

- **Alprazolam:** - Anxiolytic and Antidepressant

3. Short acting BZD:

- **Triazolam:** - Hypnotic.
- **Midazolam:** - I.V Anaesthetic.

Indications of benzodiazepines

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1. **Anxiety** (*Diazepam, Clonazepam and Alprazolam*)
2. **Panic attacks** (*Clonazepam, Alprazolam*)
3. **Alcohol Withdrawal** (*Diazepam*)
4. **Insomnia** (*Triazolam, Diazepam*)
5. **Sleep walking and Night terrors** (*Diazepam*)
6. **Facilitating Endoscopy** (*Midazolam IV, Diazepam IV*)
7. **Pre-anaesthetic medication** (*Midazolam IV, Diazepam IV*)
8. **Adjuvant to anaesthesia** (*Midazolam IV, Diazepam IV*)

Adverse effects of Benzodiazepine drugs

- 1) Drowsiness, Confusion (and lack of learning abilities in children)
- 2) Ataxia and decreased motor skills
- 3) Anterograde amnesia (*loss of memory for recent events*)
- 4) Dependence which may lead to Abuse
- 5) Addictive potentiation of CNS depression with Alcohol (and other CNS depressants)
- 6) Excessive daytime sedation (Hangover), decreased REM sleep
→ Irritability, Emotional distress and rebound insomnia (with hypnotics)

Benzodiazepine antagonist

- ▶ **Flumazenil** is a GABA receptor antagonist; rapidly reverses effects of benzodiazepines
- ▶ Flumazenil is used as an anti dote for benzodiazepines
- ▶ *Available by IV; Rapid onset of action but short duration*

Sedative-Hypnotics: **Barbiturates**

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Drug name	Clinical Uses	Common Adverse effects
Phenobarbital	Insomnia, Seizures, Convulsions, Pre-anaesthetic use	Respiratory & CNS depression, Somnolence, Confusion, ataxia, vertigo, nausea, constipation, bradycardia, hypotension
Pentobarbital	Pre-operative sedation, hypnotic	<i>Similar to Phenobarbital</i>
Thiopental	Induction of anaesthesia, Hypnotic, Convulsions, Sedation	Cardiac & Respiratory depression, arrhythmia, rectal bleeding, etc <i>C A R</i>

Sedative-Hypnotics: Other agents

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Drug name	Clinical Uses	Common Adverse effects
Zolpidem	Hypnotic	Dysphoria, Drowsiness, Amnesia, Dizziness, Nausea, vomiting, etc DNV → DA
Buspirone	Anxiety disorders	Dizziness, Headache, Chest pain, Insomnia, Thrombocytopenia
Zeleplon	Insomnia	Visual impairment, myalgia, Dizziness, Rebound insomnia & anxiety, headache, etc VH → MARD

Guidelines for use of Sedative – hypnotics Cont'd

- ❖ *Ideal hypnotic drug choice should have rapid onset and short duration of action (to avoid excessive daytime sedation next day)*
- ❖ *Longer acting drugs are preferred as Anxiolytics*
- ❖ *Dose-adjustment is required for patients with impaired liver function*
- ❖ *Dose reduction necessary in Geriatrics & Paediatrics.*

Contraindications

- ▶ Sedative-Hypnotic drugs not given to patients with known hypersensitivity to the drugs
- ▶ Avoid in comatose patients or those with respiratory problems
- ▶ Avoided in patients with history of recent chronic drug abuse (*e.g. alcohol, psychotropics, etc*)
- ▶ Avoid in Pregnancy & lactation

QUESTIONS

1. Which of the following drug is short acting benzodiazepines
 - a. Midazolam
 - b. Alprazolam
 - c. Diazepam
 - d. clonazepam

2. Which of the following barbiturate is ultra short acting and used as anaesthesia

- a. Phenobarbitone
- b. Pentobarbitone
- c. Thiopental
- d. flumazenil

END